

# Theory of Discounted Markov Decision Processes

Mohammad Sadegh Talebi

[m.shahi@di.ku.dk](mailto:m.shahi@di.ku.dk)

Department of Computer Science



# Markov Decision Process

A finite **Markov Decision Process (MDP)** is a tuple  $M = (\mathcal{S}, \mathcal{A}, P, R)$ :

- **State-space  $\mathcal{S}$**  (with size  $S$ )
- **Action-space  $\mathcal{A}$**  (with size  $A$ )
- **Transition function  $P$** : Selecting  $a \in \mathcal{A}$  in  $s \in \mathcal{S}$  leads to a transition to  $s'$  with probability  $P(s'|s, a)$ .  $P(\cdot|s, a)$  is a probability distribution over  $\mathcal{S}$ , i.e.,

$$\sum_{s' \in \mathcal{S}} P(s'|s, a) = 1$$

- **Reward function  $R$** : Selecting  $a \in \mathcal{A}$  in  $s \in \mathcal{S}$  yields a reward  $r \sim R(s, a)$ .
- The action-space may generally be state-dependent; we use  $\mathcal{A}_s$  to denote the set of actions available in state  $s$ .
- In general,  $\mathcal{S}$  or  $\mathcal{A}$  could be finite, countably infinite, or continuous.

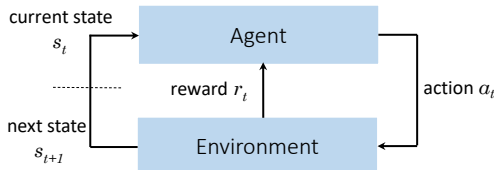


## Recap: Interaction with MDP

An **agent** interacts with the MDP for  $N$  rounds.

At each time step  $t$ :

- The agent observes the current state  $s_t$  and takes an action  $a_t \in \mathcal{A}$
- The environment (MDP) decides a reward  $r_t := r(s_t, a_t) \sim R(s_t, a_t)$  and a next state  $s_{t+1} \sim P(\cdot | s_t, a_t)$
- The agent receives  $r_t$  (any time in step  $t$  before start of  $t + 1$ )



This interaction produces a trajectory (or history)

$$h_t = (s_1, a_1, r_1, s_2, a_2, r_2, \dots, s_{t-1}, a_{t-1}, r_{t-1}, s_t)$$



## Objective Function

**Infinite-Horizon Discounted MDPs:**  $N = \infty$ , and the goal is to maximize the total expected sum of **discounted** rewards

$$\max_{\text{all strategies}} \mathbb{E} \left[ \sum_{t=1}^{\infty} \gamma^{t-1} r(s_t, a_t) \right]$$

Two views on discounting with a discount factor  $\gamma \in [0, 1)$ :

- Earlier rewards are more important. A unit reward at present will worth  $\gamma$  in the next slot.
- Problems with random horizon  $N$  and absorbing states



## Reward Function: Some Comments

- **Bounded Rewards Assumption:** We assume

$$R_{\max} := \sup_{s,a} |\mathbb{E}_{r \sim R(s,a)}[r]| < \infty$$

- For simplicity, we assume *deterministic rewards*
  - Hence,  $r \sim R(s,a)$  means  $r = R(s,a)$ .
  - Hence, we may use  $r(s,a)$  and  $R(s,a)$  interchangeably, but tend to keep  $r(s,a)$  for generality.
  - The results in this lecture will hold for stochastic rewards under mild assumptions (and often by replacing  $R(s,a)$  or  $r(s,a)$  with its mean).

**This lecture:** We consider deterministic and bounded rewards.



# Value Function



## Recap: Policy

When interacting with an MDP, actions are taken according to some **policy**:

|                   | deterministic                                                      | randomized                                                                 |
|-------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------|
| stationary        | $\pi : \mathcal{S} \rightarrow \mathcal{A}, \quad \Pi^{\text{SD}}$ | $\pi : \mathcal{S} \rightarrow \Delta(\mathcal{A}), \quad \Pi^{\text{SR}}$ |
| history-dependent | $\pi : \mathcal{H} \rightarrow \mathcal{A}, \quad \Pi^{\text{HD}}$ | $\pi : \mathcal{H} \rightarrow \Delta(\mathcal{A}), \quad \Pi^{\text{HR}}$ |

- $\Delta(\mathcal{A})$  denotes the simplex of probability distributions over  $\mathcal{A}$ .
- $\mathcal{H}$  the set of all possible histories (trajectories).

For  $\pi \in \Pi^{\text{SR}}$ , we write  $a \sim \pi(\cdot|s)$  or  $a \sim \pi(s)$ . Also, given  $f : \mathcal{A} \rightarrow \mathbb{R}$ ,

$$\mathbb{E}_{a \sim \pi(s)}[f(a)] = \sum_{a \in \mathcal{A}} f(a) \pi(a|s)$$



## Value Function

The **value function** of policy  $\pi$  (or simply, **value of  $\pi$** ) is a mapping  $V^\pi : \mathcal{S} \rightarrow \mathbb{R}$  defined as

$$V^\pi(s) := \mathbb{E}^\pi \left[ \sum_{t=1}^{\infty} \gamma^{t-1} r(s_t, a_t) \middle| s_1 = s \right].$$

where  $\mathbb{E}^\pi$  indicates expectation over trajectories generated by  $\pi$ .

- Intuitively,  $V^\pi(s)$  measures the sum of future discounted rewards (in expectation) when the agent starts in  $s$  and follows  $\pi$ .
- A rough upper bound:

$$|V^\pi(s)| \leq \frac{R_{\max}}{1 - \gamma}, \quad \forall s \in \mathcal{S}$$



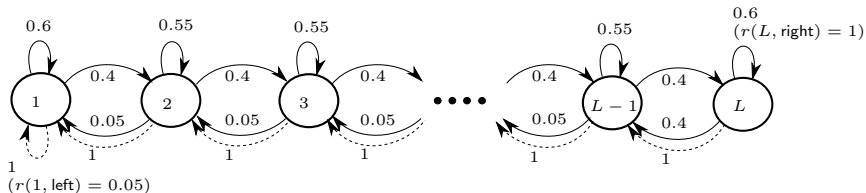
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$$V^\pi(s) := \mathbb{E}^\pi \left[ \sum_{t=1}^{\infty} \gamma^{t-1} r(s_t, a_t) \mid s_1 = s \right].$$

where  $\mathbb{E}^\pi$  indicates expectation over trajectories generated by  $\pi$ .

**Example:** Value of  $\pi = \text{'always left'}$ ?



## Value Function

The **value function** of policy  $\pi$  (or simply, **value of  $\pi$** ) is a mapping  $V^\pi : \mathcal{S} \rightarrow \mathbb{R}$  defined as

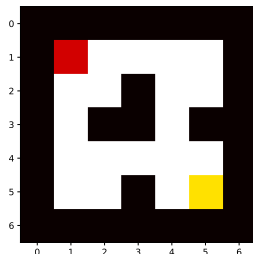
$$V^\pi(s) := \mathbb{E}^\pi \left[ \sum_{t=1}^{\infty} \gamma^{t-1} r(s_t, a_t) \middle| s_1 = s \right].$$

where  $\mathbb{E}^\pi$  indicates expectation over trajectories generated by  $\pi$ .

We may be interested in  $V^\pi(s_{\text{init}})$ .

### Example: 4-room Grid-World

- $s_{\text{init}}$  is ■.
- Terminal state ■.
- Our interest is to compute/estimate  $V^\pi(\text{■})$



## Action-Value Function

The **action-value function** of policy  $\pi$  (or simply, **Q-value of  $\pi$** ) is a mapping  $Q^\pi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$  defined as (Under the bounded reward assumption)

$$Q^\pi(s, a) := \mathbb{E}^\pi \left[ \sum_{t=1}^{\infty} \gamma^{t-1} r(s_t, a_t) \middle| s_1 = s, a_1 = a \right].$$

- Intuitively,  $Q^\pi(s, a)$  measures the sum of future discounted rewards (in expectation) when the agent starts in  $s$  and takes action  $a$  in the first step (possibly  $a \neq \pi(s)$ ), and then follows  $\pi$  afterwards.
- A rough upper bound:

$$|Q^\pi(s, a)| \leq \frac{R_{\max}}{1 - \gamma}, \quad \forall s \in \mathcal{S}, \forall a \in \mathcal{A}$$

- For all  $s \in \mathcal{S}$ ,  $Q^\pi(s, \pi(s)) = V^\pi(s)$ .



# Policy Evaluation



## Recap: Induced Markov Chains

- Every  $\pi \in \Pi^{\text{SR}}$  induces a **Markov chain** on  $M$ , with transition probability matrix  $P^\pi$  given by:

$$P_{s,s'}^\pi = \sum_{a \in \mathcal{A}} P(s'|s, a) \pi(a|s), \quad s, s' \in \mathcal{S}.$$

- Every  $\pi \in \Pi^{\text{SR}}$  induces a reward vector  $r^\pi \in \mathbb{R}^{\mathcal{S}}$  on  $M$  defined by:

$$r^\pi(s) = \sum_{a \in \mathcal{A}} R(s, a) \pi(a|s), \quad s \in \mathcal{S}.$$

- If  $\pi \in \Pi^{\text{SD}}$ , then  $P_{s,s'}^\pi = P(s'|s, \pi(s))$  and  $r^\pi(s) = R(s, \pi(s))$ .

Every policy  $\pi \in \Pi^{\text{SR}}$  induces a **Markov Reward Process (MRP)** on  $M$ , specified by  $r^\pi$  and  $P^\pi$ .



## Bellman Equation for $\pi$

### Theorem (Bellman Equation for $\pi$ )

Let  $\pi \in \Pi^{SR}$ . For all  $s \in \mathcal{S}$ ,

$$\begin{aligned} V^\pi(s) &= \mathbb{E}_{a \sim \pi(s)}[r(s, a)] + \gamma \mathbb{E}_{a \sim \pi(s)} \left[ \sum_{x \in \mathcal{S}} P(x|s, a) V^\pi(x) \right] \\ &= \sum_{a \in \mathcal{A}} \pi(a|s) r(s, a) + \gamma \sum_{a \in \mathcal{A}} \pi(a|s) \sum_{x \in \mathcal{S}} P(x|s, a) V^\pi(x) \end{aligned}$$

Equivalently,  $V^\pi = r^\pi + \gamma P^\pi V^\pi$ .

- These relations are called the **Bellman equation**.
- The theorem tells us that for  $\pi \in \Pi^{SR}$ ,  $V^\pi$  satisfies the Bellman equation.
- For a deterministic policy  $\pi \in \Pi^{SD}$ , the Bellman equation becomes:

$$V^\pi(s) = r(s, \pi(s)) + \gamma \sum_{x \in \mathcal{S}} P(x|s, \pi(s)) V^\pi(x), \quad s \in \mathcal{S}.$$



## Bellman Operator for $\pi$

The **Bellman operator** associated to  $\pi \in \Pi^{\text{SR}}$  is a mapping  $\mathcal{T}^\pi : \mathbb{R}^S \rightarrow \mathbb{R}^S$ , such that for any function  $f : S \rightarrow \mathbb{R}$ ,

$$\mathcal{T}^\pi f := r^\pi + \gamma P^\pi f.$$

- Intuitively,  $\mathcal{T}^\pi$  is the value of  $\pi$  for the same one-stage problem.
- $\mathcal{T}^\pi$  applies to (or *operates on*) a function defined on  $S$  and returns another function defined on  $S$ .
- The Bellman equation  $V^\pi = r^\pi + \gamma P^\pi V^\pi$  reads

$$V^\pi = \mathcal{T}^\pi V^\pi$$

In other words,  $V^\pi$  is the *unique* fixed-point of the operator  $\mathcal{T}^\pi$ .



## Bellman Equation for $\pi$

We prove the theorem for  $\pi \in \Pi^{\text{SD}}$ . (See Lecture Notes for  $\pi \in \Pi^{\text{SR}}$ .)

**Proof.** Let  $\pi \in \Pi^{\text{SD}}$  and  $s \in \mathcal{S}$ . We have

$$\begin{aligned}
 V^\pi(s) &= \mathbb{E}^\pi \left[ \sum_{t=1}^{\infty} \gamma^{t-1} r(s_t, \pi(s_t)) \middle| s_1 = s \right] \\
 &= r(s, \pi(s)) + \mathbb{E}^\pi \left[ \sum_{t=2}^{\infty} \gamma^{t-1} r(s_t, \pi(s_t)) \middle| s_1 = s \right] \\
 &= r(s, \pi(s)) + \gamma \sum_{x \in \mathcal{S}} \mathbb{P}(s_2 = x | s_1 = s, a_1 = \pi(s_1)) \underbrace{\mathbb{E}^\pi \left[ \sum_{t=2}^{\infty} \gamma^{t-2} r(s_t, \pi(s_t)) \middle| s_2 = x \right]}_{= V^\pi(x)} \\
 &= r(s, \pi(s)) + \gamma \sum_{x \in \mathcal{S}} \mathbb{P}(s_2 = x | s_1 = s, a_1 = \pi(s_1)) V^\pi(x) \\
 &= r(s, \pi(s)) + \gamma \sum_{x \in \mathcal{S}} P(x | s, \pi(s)) V^\pi(x).
 \end{aligned}$$



# Policy Evaluation

**Policy Evaluation:** Computing  $V^\pi$  for a given  $\pi$

- **Direct Computation:** Using Bellman equation,

$$V^\pi = r^\pi + \gamma P^\pi V^\pi \implies I - \gamma P^\pi \text{ is invertible} \quad V^\pi = (I - \gamma P^\pi)^{-1} r^\pi$$

- **Iterative Policy Evaluation:** Using  $V^\pi = \mathcal{T}^\pi V^\pi$ , the sequence

$$V_{n+1} = \mathcal{T}^\pi V_n = \underbrace{\mathcal{T}^\pi \cdots \mathcal{T}^\pi}_{n+1 \text{ times}} V_0$$

converges to  $V^\pi$  starting from any  $V_0$ .

- **Monte-Carlo Method:** Generate a number of trajectories of  $\pi$  and use the sample mean as an estimator to  $V^\pi$ .



So far:

- We defined policies and the value function.
- We characterized the value of stationary policies (via Bellman equations and operator).
- We developed ways to compute the value of a *fixed* stationary policy.

*How to find an optimal strategy/policy? Alternatively, how to find policies with good values?*



# Optimization in Discounted MDPs: Optimal Policy and Value



## Optimal Value and Policy

Solving a discounted MDP  $M$  amounts to solving the following optimization problem:

$$V^*(s) = \sup_{\pi \in \Pi^{\text{HR}}} V^\pi(s), \quad \forall s \in \mathcal{S}.$$

- (i)  $V^* : \mathcal{S} \rightarrow \mathbb{R}$  is called the **optimal value** function.
- (ii) If there exists  $\pi^*$  such that  $V^{\pi^*}(s) = V^*(s)$  for all  $s \in \mathcal{S}$ , then  $\pi^*$  is called an **optimal policy**.
- (iii)  $\pi$  is  **$\varepsilon$ -optimal** for  $\varepsilon > 0$  if

$$V^\pi(s) \geq V^*(s) - \varepsilon, \quad \forall s \in \mathcal{S}$$



# Bellman Optimality Equation

## Theorem

$V^*$  satisfies the *optimal Bellman equation*:

$$V^*(s) = \max_{a \in \mathcal{A}} \left( r(s, a) + \gamma \sum_{x \in \mathcal{S}} P(x|s, a) V^*(x) \right), \quad s \in \mathcal{S}$$

The *optimal Bellman operator* is a mapping  $\mathcal{T} : \mathbb{R}^{\mathcal{S}} \rightarrow \mathbb{R}^{\mathcal{S}}$ , such that for any function  $f : \mathcal{S} \rightarrow \mathbb{R}$ ,

$$(\mathcal{T}f)(s) := \max_{a \in \mathcal{A}} \left( r(s, a) + \gamma \sum_{x \in \mathcal{S}} P(x|s, a) f(x) \right), \quad s \in \mathcal{S}$$

- $V^*$  satisfies  $\mathcal{T}V^* = V^*$ .
- We can define  $\mathcal{T}$  and optimal Bellman equation for the optimal Q function (next lecture).



# Optimality Theorems

## Theorem

*Suppose the state space  $\mathcal{S}$  is finite. Then there exists a policy  $\pi^* \in \Pi^{SD}$ .*

- Thus, when seeking  $\pi^*$  in a discounted MDP with a finite  $\mathcal{S}$ , we can restrict our attention to  $\Pi^{SD}$ .
- In other words, for finite  $\mathcal{S}$ ,

$$\sup_{\pi \in \Pi^{HR}} V^\pi = \sup_{\pi \in \Pi^{SD}} V^\pi = \max_{\pi \in \Pi^{SD}} V^\pi$$



# Optimality Theorems

A fundamental result in the theory of discounted MDPs:

## Theorem

A stationary deterministic policy  $\pi$  is optimal *if and only if*

$$\mathcal{T}^\pi V^* = \mathcal{T}V^*$$

Equivalently,  $\pi$  is optimal *if and only if* it attains the maximum in the Bellman optimality equations: For all  $s \in \mathcal{S}$ ,

$$\pi(s) \in \operatorname{argmax}_{a \in \mathcal{A}} \left( r(s, a) + \gamma \sum_{x \in \mathcal{S}} P(x|s, a) V^*(x) \right).$$



So far:

- We defined policies and the value function.
- We characterized the value of stationary policies (via Bellman equations and operator).
- We developed ways to compute the value of a *fixed* stationary policy.
- We defined the notion of optimality and showed that *there exists*  $\pi^* \in \Pi^{\text{SD}}$  when  $\mathcal{S}$  is finite.
- We characterized the optimal value function  $V^*$  (via *optimal Bellman equation*).

*How to actually compute  $\pi^*$ ?*



# Algorithms for Solving Discounted MDPs



# Major Solution Methods

Three major classes of algorithms for solving discounted MDPs:

- Value Iteration
- Policy Iteration
- Linear Programming



## Value Iteration

### Value Iteration (VI)

- The most well-known, and perhaps the simplest, algorithm for solving discounted MDPs
- Around since the early days of MDPs
- Also known as successive approximation, backward induction, etc.

**Idea:** The optimal Bellman operator  $\mathcal{T}$  is *contracting*. Iterate  $\mathcal{T}$  until convergence:

$$V_{n+1} = \mathcal{T}V_n, \quad n = 0, 1, 2, \dots$$

Indeed, VI is an algorithm for approximating the fixed point of  $\mathcal{T}$ .



## Value Iteration (VI)

**input:**  $\varepsilon$

- **initialization:** Select  $V_0 \in \mathbb{R}^S$  arbitrarily. Set  $n = 0$ .
- **repeat:**
  - (ii)  $n = n + 1$ .
  - (i) Update, for each  $s \in S$ ,

$$V_n(s) = \max_{a \in \mathcal{A}} \left( r(s, a) + \gamma \sum_{x \in S} P(x|s, a) V_{n-1}(x) \right)$$

**until**  $(\|V_n - V_{n-1}\|_\infty \leq \frac{\varepsilon(1-\gamma)}{2\gamma})$

**output:**

$$\pi^{\text{VI}}(s) \in \arg \max_{a \in \mathcal{A}} \left( r(s, a) + \gamma \sum_{x \in S} P(x|s, a) V_n(x) \right), \quad s \in S$$



# Why VI works?

*Why does VI work?*

⇒ Because of contraction properties of Bellman operators.



## Contraction Mapping

An operator (or mapping)  $\mathcal{L} : \mathbb{R}^n \rightarrow \mathbb{R}^n$  is called a  **$\kappa$ -contraction mapping** (with respect to  $\|\cdot\|$ ) if there exists  $\kappa \in [0, 1)$  such that for all  $v, v' \in \mathbb{R}^n$ ,

$$\|\mathcal{L}v - \mathcal{L}v'\| \leq \kappa \|v - v'\|.$$

### Theorem (Banach Fixed-Point Theorem)

Suppose  $\mathcal{L}$  is a contraction mapping. Then

- (i) *there exists a unique  $v^* \in \mathbb{R}^n$  such that  $\mathcal{L}v^* = v^*$ ;*
- (ii) *for any  $v_0 \in \mathbb{R}^n$ , the sequence  $(v_n)_{n \geq 0}$  with  $v_{n+1} = \mathcal{L}v_n = \mathcal{L}^{n+1}v_0$  for  $n \geq 0$  converges to  $v^*$ .*



# $\mathcal{T}^\pi$ and $\mathcal{T}$ Are Contraction Mapping

## Lemma

For any  $v, v' \in \mathbb{R}^S$ , and any  $\pi$ ,

$$\|\mathcal{T}^\pi v - \mathcal{T}^\pi v'\|_\infty \leq \gamma \|v - v'\|_\infty,$$

$$\|\mathcal{T} v - \mathcal{T} v'\|_\infty \leq \gamma \|v - v'\|_\infty.$$

Hence,  $\mathcal{T}^\pi$  and  $\mathcal{T}$  are  $\gamma$ -contraction mappings w.r.t.  $\|\cdot\|_\infty$ .

**Proof.** First statement is easy to prove. For the second, we have:

$$\begin{aligned} & \|\mathcal{T} v - \mathcal{T} v'\|_\infty \\ &= \max_s \left| \max_{a \in \mathcal{A}} \left( r(s, a) + \gamma \sum_j P(j|s, a) v(j) \right) - \max_{a \in \mathcal{A}} \left( r(s, a) + \gamma \sum_j P(j|s, a) v'(j) \right) \right| \\ &\leq \max_s \max_{a \in \mathcal{A}} \left| \gamma \sum_j P(j|s, a) (v(j) - v'(j)) \right| \\ &\quad \text{(Using inequality } |\max_x f(x) - \max_x g(x)| \leq \max_x |f(x) - g(x)|) \\ &\leq \gamma \max_s \max_{a \in \mathcal{A}} \max_j |v(j) - v'(j)| \sum_j P(j|s, a) = \gamma \|v - v'\|_\infty \end{aligned}$$



## VI: Convergence

VI is a *globally convergent* method for finding an  $\varepsilon$ -optimal policy. Formally:

### Theorem

Let  $(V_n)_{n \geq 0}$  a sequence of value functions generated by VI with some  $\varepsilon > 0$  starting from an arbitrary initial point  $V_0 \in \mathbb{R}^S$ . Then,

- (i)  $V_n$  converges to  $V^*$  in norm;
- (ii) the algorithm stops after finitely many iterations;
- (iii)  $\pi^{VI}$  is  $\varepsilon$ -optimal;
- (iv) when convergence criterion is satisfied,  $\|V_{n+1} - V^*\|_\infty < \varepsilon/2$ .

- Each iteration of VI involves  $O(S^2 A)$  arithmetic calculations.
- The iteration complexity of VI depends on both  $\varepsilon$  and  $\gamma$ . The larger the  $\gamma$ , the more iteration until the algorithm finds an  $\varepsilon$ -optimal policy.



# Policy Iteration

## Policy Iteration (PI)

- A popular algorithm for solving discounted MDPs
- Around since early days of MDPs
- Like VI, it is an iterative algorithm but directly searches in the space of policies.

**Idea:** Starting from an initial policy, at each iterate  $n$ ,

- Find  $V^{\pi_n}$  (policy evaluation)
- Improve  $\pi_n$  to  $\pi_{n+1}$  using  $V^{\pi_n}$  (policy improvement)



## Policy Iteration (PI)

- **initialization:** Select  $\pi_0$  and set  $n = 0$
- **repeat**

(i) *Policy Evaluation:* Find  $V_n$ , the value of  $\pi_n$  by solving

$$(I - \gamma P^{\pi_n})V_n = r^{\pi_n}$$

(ii) *Policy Improvement:* Choose  $\pi_{n+1}$  such that

$$\pi_{n+1}(s) \in \operatorname{argmax}_{a \in \mathcal{A}} \left( r(s, a) + \gamma \sum_{x \in \mathcal{S}} P(x|s, a) V_n(x) \right)$$

and if possible, set  $\pi_{n+1} = \pi_n$ .

(iii)  $n = n + 1$ .

- **until**  $(\pi_{n+1} = \pi_n)$
- **output:**  $\pi^{\text{PI}} = \pi_n$



## PI: Convergence

### Theorem

*Suppose  $M$  has a finite state-action space. Then,*

*(i) PI terminates in at most*

$$\min \left\{ O\left(\frac{A^S}{S}\right), O\left(\frac{SA}{1-\gamma} \log \frac{1}{1-\gamma}\right) \right\} \quad \text{iterations;}$$

*(ii)  $\pi^{PI} = \pi^*$ .*

- Under PI,  $V_{n+1} \geq V_n$  for any  $n$ . Further, the number of policies is finite  $A^S$ .
- Each iteration in PI involves solving a linear system with  $S$  equations and  $S$  unknowns. Hence, per iteration complexity of PI is  $O(S^3 + S^2 A)$ .
- In practice, PI converges within, at most, a few tens of iterations.



## Reward Function: Some Comments

Two Reward Models:

- $R(s, a)$ : Reward distribution in state  $s$  when executing action  $a$
- $R(s, a, s')$ : Reward distribution in state  $s$  when executing action  $a$  and the next state is  $s'$

We consider the first model, but the two models are related:

$$R(s, a) = \sum_{s' \in \mathcal{S}} R(s, a, s') P(s' | s, a)$$

